

REQ-2: Non-Functional Requirements Traced to ISO 25010 Maintainability Sub-Characteristics

Project: Software Engineering Coursework
Task ID: REQ-2
Owner: POL
Sprint Days: 3–4
Feeds into: Requirements Document, T4 (Test Plan), LO2

Overview

This document captures the Non-Functional Requirements (NFRs) for the system, each traced to a specific sub-characteristic of the ISO/IEC 25010 Maintainability quality characteristic. Every NFR includes a measurable acceptance criterion to satisfy the Definition of Done (DoD).

ISO 25010 defines five Maintainability sub-characteristics:

Sub-Characteristic	Definition
Modularity	The system is composed of discrete components such that a change to one component has minimal impact on others
Reusability	An asset can be used in more than one system or in building other assets
Analysability	The ease with which the impact of an intended change on the system can be assessed, or a cause of failure diagnosed
Modifiability	The degree to which a product or system can be effectively and efficiently modified without introducing defects
Testability	The degree of effectiveness and efficiency with which test criteria can be established and tests performed

NFR Table

NFR ID	NFR Description	ISO 25010 Sub-Characteristic	Priority	Acceptance Criterion
NFR-M1	Each system module must encapsulate a single, clearly defined responsibility	Modularity	High	No module exceeds 200 lines of code; each module has exactly one stated responsibility documented in its header comment, verified during code review

NFR ID	NFR Description	ISO 25010 Sub-Characteristic	Priority	Acceptance Criterion
NFR-M2	Changes to one module must not require modifications to more than one other unrelated module	Modularity	High	Impact analysis during sprint review confirms that any single-module change propagates to ≤ 1 other module, verified across at least 3 change scenarios
NFR-M3	Common utility functions (e.g., validation, formatting, logging) must be implemented in shared libraries and reused across modules	Reusability	Medium	Static analysis confirms zero duplication of utility logic across modules; all reusable components are stored in a dedicated <code>/utils</code> or equivalent package
NFR-M4	A developer unfamiliar with the codebase must be able to identify the root cause of a reported defect within a defined time limit	Analysability	High	In a peer review exercise, a team member not involved in the component's implementation localises the fault within 30 minutes using only the codebase and inline documentation
NFR-M5	All public methods and classes must include inline documentation describing their purpose, parameters, and return values	Analysability	Medium	Documentation coverage $\geq 90\%$ of all public interfaces, verified by automated documentation tool (e.g., Javadoc, Doxygen, or equivalent)
NFR-M6	Implementing a new feature of defined scope must not require modification of more than three existing source files	Modifiability	High	Verified by tracking file-change diffs in version control during at least two feature additions in the sprint; average files changed ≤ 3
NFR-M7	Introducing a modification must not cause regressions in existing functionality	Modifiability	High	All existing automated tests pass after any modification; zero regression failures in the CI/CD pipeline
NFR-M8	All business logic functions must be independently unit-testable in isolation from external dependencies	Testability	High	$\geq 80\%$ unit test code coverage achieved across all business logic modules, measured by an automated coverage tool (e.g., JaCoCo, Coverage.py); dependency injection or mocking used where required
NFR-M9	The system must support automated regression testing via a CI pipeline	Testability	High	A CI pipeline (e.g., GitHub Actions) runs the full test suite on every commit; test results and coverage reports are generated automatically and accessible to all team members
NFR-M10	Test data must be separate from production logic and configurable without code changes	Testability	Medium	All test data is stored in external configuration files or test fixtures; no hardcoded test values appear in production source code, verified by code review

Traceability Summary

ISO 25010 Sub-Characteristic	Covered By
Modularity	NFR-M1, NFR-M2
Reusability	NFR-M3
Analysability	NFR-M4, NFR-M5
Modifiability	NFR-M6, NFR-M7
Testability	NFR-M8, NFR-M9, NFR-M10

Notes

- All five ISO 25010 Maintainability sub-characteristics are covered by at least one NFR.
- Each acceptance criterion is measurable and verifiable — either through automated tooling, code review, or a defined peer exercise.
- This table feeds directly into:
 - Requirements Document — NFR section
 - T4 — Test plan (NFR-M8, NFR-M9, NFR-M10 directly inform test strategy and coverage targets)
 - LO2 — Demonstrates ability to elicit and specify verifiable non-functional requirements aligned to a recognised quality standard